Development Environment Setup

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If using the factory image, the development environment is pre-installed; please skip the installation steps.

I. ComfyUI Development Environment Setup

Note: The following installation process must be performed in the same terminal.

1. Install ComfyUI

Download ComfyUI source code

```
cd ~/
git clone https://github.com/Comfy-Org/ComfyUI.git
```

Create a new venv virtual environment

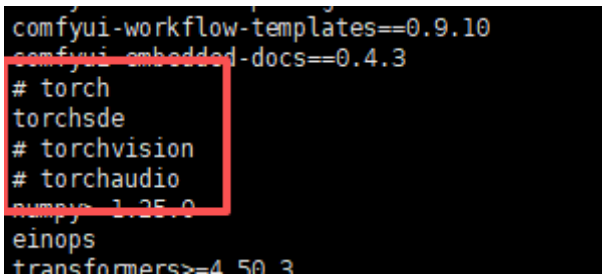
```
cd ~/ComfyUI
python -m venv venv
```

Activate the virtual environment

```
source ~/ComfyUI/venv/bin/activate
```

Modify the environment dependency file, commenting out torch, torchvision, and torchaudio.

```
vim ~/ComfyUI/requirements.txt
```



Then install the environment dependencies.

```
pip install -r requirements.txt
```

Install torch, torchvision, and torchaudio from jetson-ai-lab.io

```
pip install --pre torch, torchvision, torchaudio --index-url https://pypi.jetson-ai-lab.io/sbsa/cu130
```

2. Start ComfyUI

Open the terminal and activate the virtual environment

```
source ~/ComfyUI/venv/bin/activate
```

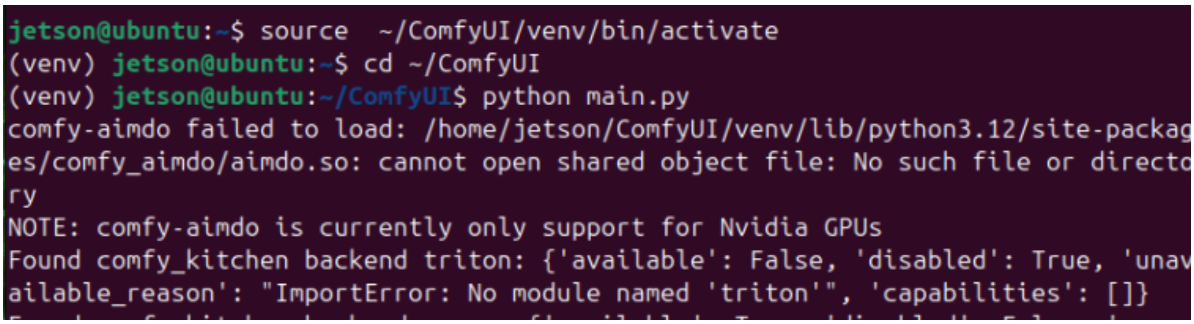
Due to network issues, accessing the Huggingface website may be inaccessible in China. Adding the HF_ENDPOINT variable can speed up access.

Run the following command in your terminal to add the `HF_ENDPOINT` variable (optional):

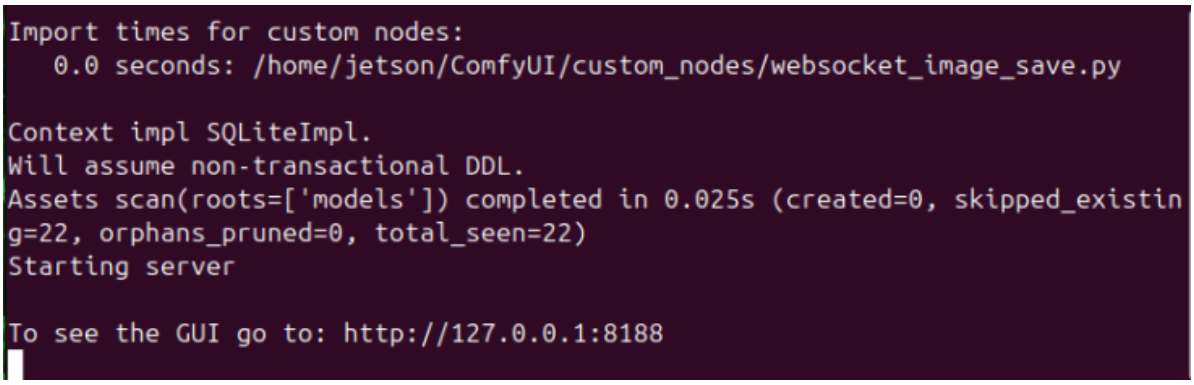
```
export HF_ENDPOINT=https://hf-mirror.com
```

Run the following commands to launch the ComfyUI application:

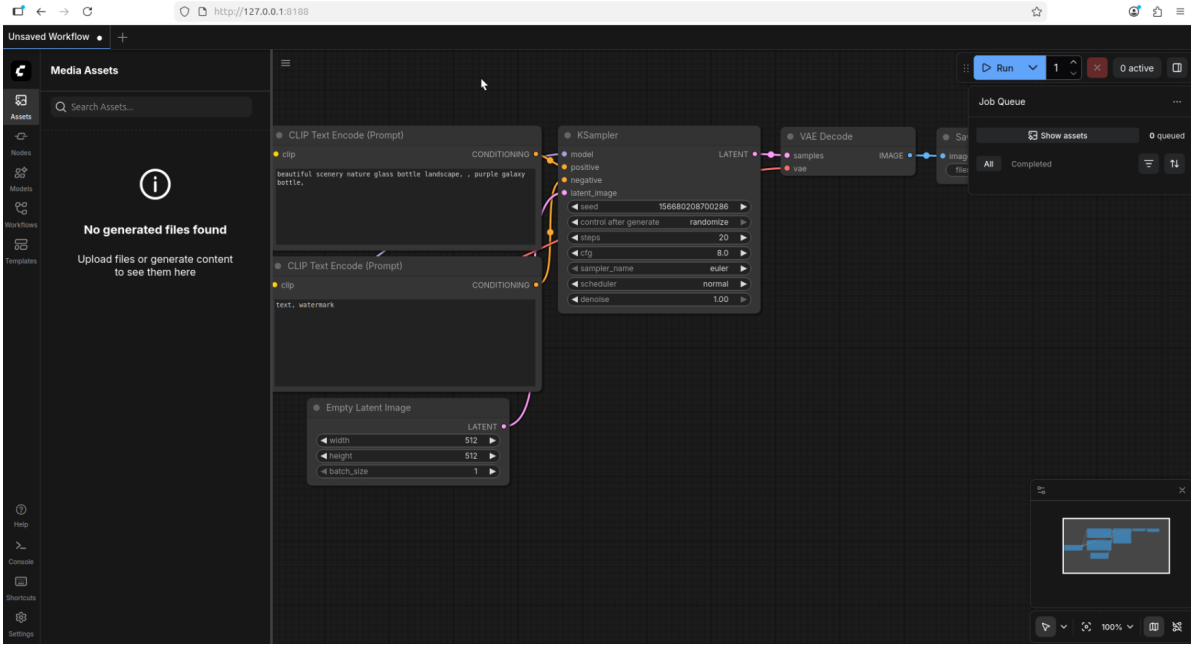
```
cd ~/ComfyUI
python main.py
```



If the application launches successfully, the ComfyUI URL will be printed to the console: `http://127.0.0.1:8188`

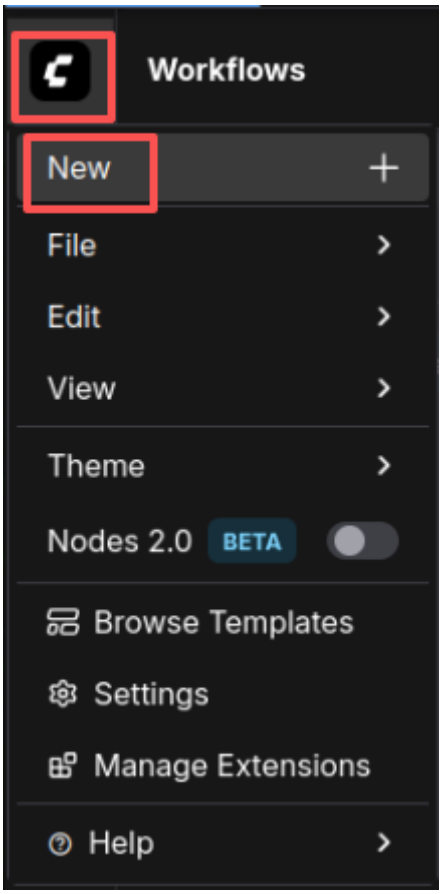


Open the ComfyUI URL in your system's web browser.

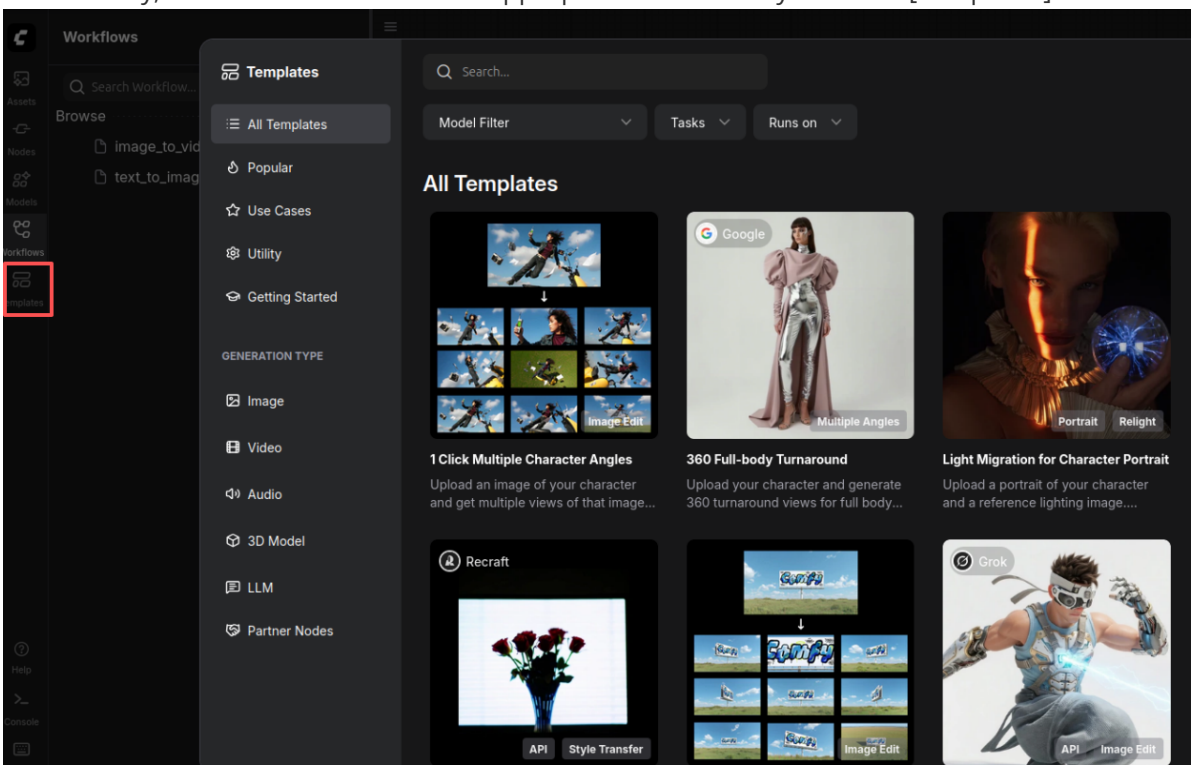


3. Exporting the ComfyUI API JSON File

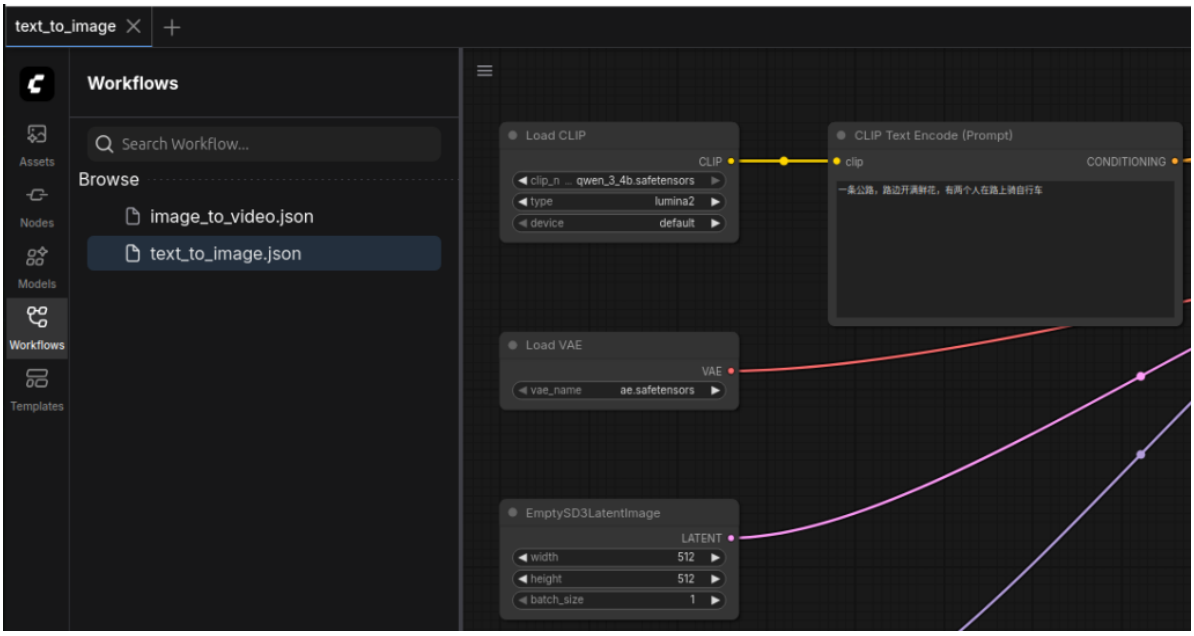
Click the menu icon in the top-left corner, then click [New] to create a blank workflow. Design your workflow according to your specific needs, and then save it.



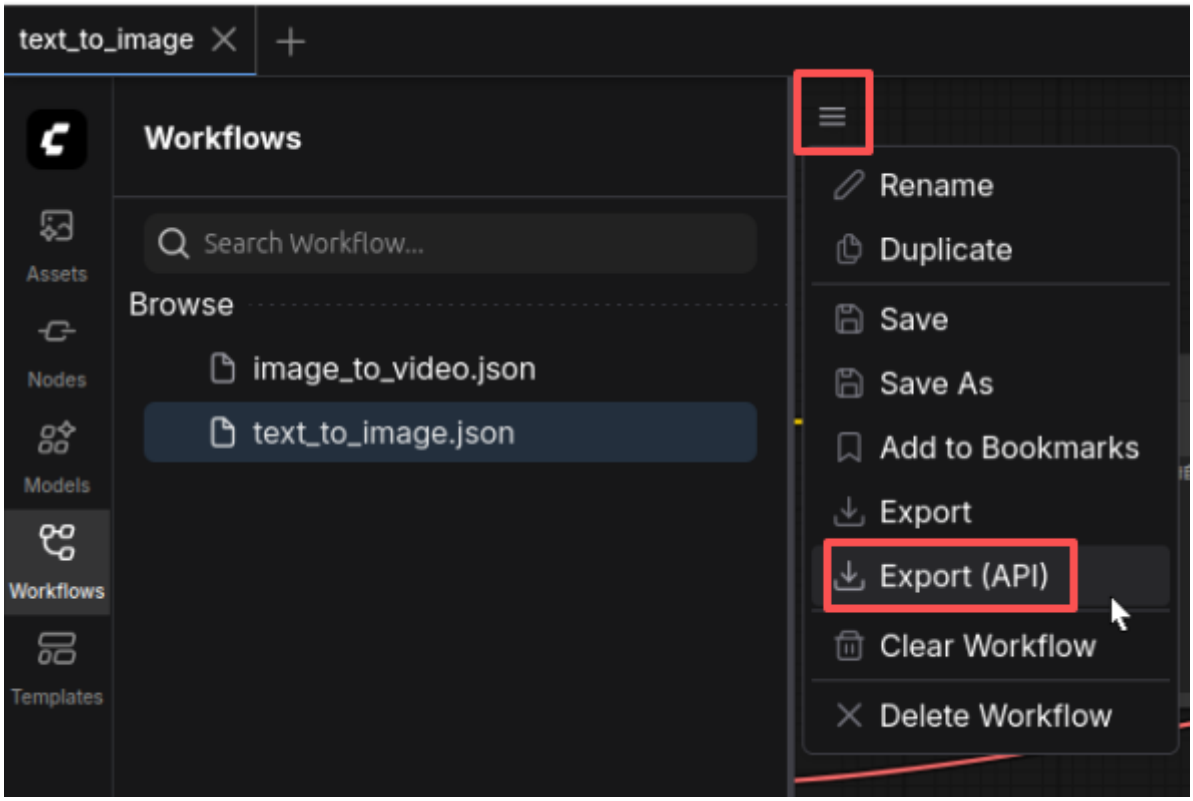
Alternatively, select a workflow with the appropriate functionality from the [Templates] section.



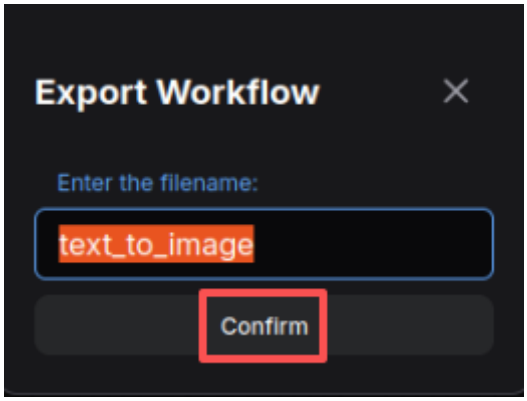
Using the text-to-image workflow as an example here: click on [Workflows], then select [text_to_image.json].



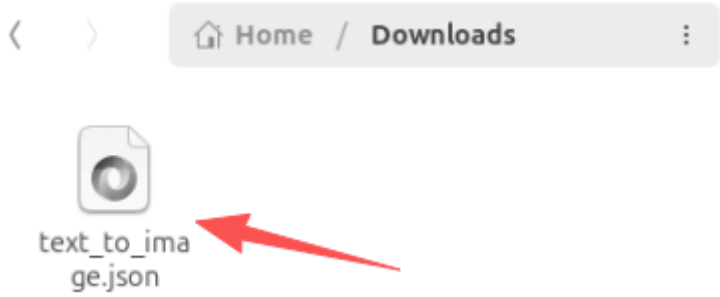
Click the three-line menu button, then select [Export (API)].



Click [Confirm].



You can find the JSON file in your system's [Downloads] directory.



II. Setting Up the Workspace Virtual Environment

Note: Open a new terminal window; all of the following installation commands should be executed within this same terminal session. ### 1. Set Up Virtual Environment

Open the terminal and create a new virtual environment for your workspace:

```
cd ~/showing
python -m venv venvs
```

Activate the virtual environment:

```
source ~/showing/venvs/bin/activate
```

Install dependencies:

```
sudo apt install libgtk2.0-dev
sudo apt install portaudio19-dev
sudo apt install screen
sudo apt install ffmpeg
sudo apt install ninja-build
sudo apt install libopenblas-dev
sudo apt install alsa-utils curl jq
```

```
pip install torch torchvision torchaudio --index-url https://pypi.jetson-ai-
lab.io/sbsa/cu130

pip install soundfile
pip install screeninfo
pip install websocket-client
pip install flask
pip install openai-whisper sounddevice numpy
pip install transformers
pip install opencv-python-headless
pip install soundfile
pip install opencv-python pillow websocket-client requests pyserial numpy
pip install wheel setuptools pybind11 packaging
pip install pathvalidate
pip install unicode_rbnf
```

2. Install Speech-to-Text Tools

Download the source code:

```
cd ~/showing
git clone --recursive https://github.com/OpenNMT/CTranslate2.git
cd CTranslate2
```

Compile the source code:

```
mkdir build && cd build
cmake .. \
  -DCMAKE_BUILD_TYPE=Release \
  -DWITH_CUDA=ON \
  -DWITH_CUDNN=ON \
  -DWITH_MKL=OFF \
  -DWITH_OPENBLAS=ON \
  -DCMAKE_CUDA_ARCHITECTURES=110 \
  -DBUILD_CLI=OFF \
  -DOPENMP_RUNTIME=COMP \
  -G Ninja
```

```
ninja -j"$(nproc)"
cmake -DCMAKE_INSTALL_PREFIX=$HOME/.local .
ninja install
```

Install the Python Driver Library

```
export CPLUS_INCLUDE_PATH=$HOME/.local/include:${CPLUS_INCLUDE_PATH}
export LIBRARY_PATH=$HOME/.local/lib:${LIBRARY_PATH}
export LD_LIBRARY_PATH=$HOME/.local/lib:${LD_LIBRARY_PATH}
export CMAKE_PREFIX_PATH=$HOME/.local:${CMAKE_PREFIX_PATH}

cd ../python
python setup.py bdist_wheel
pip install dist/*.whl

pip install faster-whisper
```

3. Install Text-to-Speech Tools

Install Tools

```
pip install g2pw sentence_stream
pip install piper-tts==1.4.1
```

Speech Model Download Link:

```
https://huggingface.co/rhasspy/piper-voices/tree/main
```